Algebra 2 Workbook

krista king
MATH

POWERS OF NEGATIVE BASES

■ 1. Simplify the expression.

$$-2^2$$

■ 2. Simplify the expression.

$$(-3)^4$$

■ 3. Simplify the expression.

$$(-5)^3$$

■ 4. Simplify the expression.

$$-3^3 + (-1)^5 \cdot 9^2$$

■ 5. Simplify the expression.

$$-4^2 \cdot (-2)^3 + 0^4$$

■ 6. Simplify the expression.

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$$-9^2 + (-1)^6 - 3^2 \cdot (-5)^3$$

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POWERS OF FRACTIONS

1. Simplify the expression.

$$\left(\frac{5}{6}\right)^2$$

2. Simplify the expression.

$$\left(\frac{1}{2}\right)^3 \cdot \left(\frac{2}{3}\right)^2$$

3. Simplify the expression.

$$\left(\frac{x^3}{y^5}\right)^2 \cdot \left(\frac{xy}{z^2}\right)^4$$

4. Simplify the expression.

$$\left(\frac{2}{3}\right)^4$$

5. Simplify the expression.

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$$\left(\frac{x^3}{y^2}\right)^5$$

6. Simplify the expression.

$$\left(\frac{a^2}{b}\right)^4$$

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ZERO AS AN EXPONENT

■ 1. Simplify the expression.

$$\frac{4^0 \cdot 9^2}{(-2)^0 + 2^1}$$

■ 2. Simplify the expression.

$$1{,}042^0$$

■ 3. Simplify the expression.

$$(10^2 + 3^3)^0$$

■ 4. Simplify the expression.

$$(-1)^0$$

■ 5. Simplify the expression.

$$(2ac - 4x)^0$$

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6. Simplify the expression.

$$(-100b)^0$$

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NEGATIVE EXPONENTS

■ 1. Simplify the expression.

$$3 \cdot 5^{-2} \cdot 6^{-2}$$

■ 2. Simplify the expression.

$$4^{-3}$$

■ 3. Simplify the expression.

$$-3^{-1}$$

■ 4. Simplify the expression.

$$-2^{-2} \cdot (-2)^{-2}$$

■ 5. Write the expression with only positive exponents.

$$a^{-5}$$

■ 6. Write the expression with only positive exponents.

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$$\frac{x^{-3}y^2}{x^4y^7}$$

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NEGATIVE EXPONENTS AND PRODUCT RULE

1. Write the expression without any negative exponents.

$$\frac{(2ab)^{-2}a^2}{b^{-4}\cdot(ab)^0}$$

2.Write the expression without any negative exponents.

$$\frac{2x^0y^{-5}}{z^{-1}(xy^2)^{-3}}$$

3. Write the expression without any negative exponents.

$$\frac{1}{a^{-8}}$$

4. Write the expression without any negative exponents.

$$\frac{8}{z^{-3}}$$

5. Write the expression without any negative exponents.

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$$\frac{2y^{-4}}{x^{-9}}$$

6. Write the expression without any negative exponents.

$$\frac{1}{(3x^{-4}y^2)^{-3}}$$

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FRACTIONAL EXPONENTS

■ 1. Simplify the expression.

$$b^2 \cdot b^{\frac{2}{3}}$$

■ 2. Simplify the expression.

$$x^5 \cdot x^{\frac{1}{6}}$$

■ 3. Simplify the expression.

$$\left(\frac{1}{16}\right)^{\frac{3}{2}}$$

■ 4. Simplify the expression.

$$8^{\frac{2}{3}}$$

■ 5. Simplify the expression.

$$3^{-\frac{3}{7}}$$

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6. Simplify the expression.

$$(81a^4b^{\frac{1}{2}})^{-\frac{5}{4}}$$

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RATIONALIZING THE DENOMINATOR

■ 1. Rationalize the denominator.

$$\frac{2}{\sqrt{5}}$$

■ 2. Rationalize the denominator.

$$\frac{1}{4\sqrt{3}}$$

■ 3. Simplify the expression, making sure to rationalize the denominator.

$$\sqrt{\frac{4}{12}}+\sqrt{\frac{9}{12}}$$

■ 4. Simplify the expression, making sure to rationalize the denominator.

$$\sqrt{\frac{6}{25}}+\sqrt{\frac{20}{24}}$$

■ 5. Simplify the expression, making sure to rationalize the denominator.

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$$\sqrt[4]{\frac{2}{3}} - \sqrt[7]{\frac{3}{2}} + \sqrt{96}$$

6. Simplify the expression, making sure to rationalize the denominator.

$$\sqrt[5]{\frac{5}{7}} + \sqrt{\frac{7}{5}} - \sqrt{140}$$

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RATIONALIZING WITH CONJUGATE METHOD

■ 1. Simplify the expression.

$$\frac{2-\sqrt{5}}{\sqrt{5}-7}$$

■ 2. Simplify the expression.

$$\frac{\sqrt{3}+\sqrt{6}}{\sqrt{6}-\sqrt{3}}$$

■ 3. Simplify the expression.

$$\frac{8}{4+\sqrt{2}}$$

■ 4. Simplify the expression.

$$\frac{x+\sqrt{5}}{-5\sqrt{x}+\sqrt{5}}$$

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5. Simplify the expression.

$$\frac{1+\sqrt{y}}{\sqrt{y}+\sqrt{3}}$$

6. Simplify the expression.

$$\frac{\sqrt{x}}{\sqrt{x}+\sqrt{y}}$$

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RATIOS AND PROPORTIONS

■ 1. The class has 12 girls and 18 boys. What is the ratio of boys to the total number of students in the class?

■ 2. The ratio of boys to girls in the class is $4 : 3$. The total number of students in the class is 28. How many girls are in the class?

■ 3. The ratio of boys to girls in the class is $5 : 3$. The total number of students in the class is 32. How many boys are in the class?

■ 4. Two numbers have a ratio of 1 to 4 and a sum of 40. What are the two numbers?

■ 5. There are 11 quarters, 9 dimes, and 13 nickels. What is the ratio of dimes to the total number of coins?

■ 6. The ratio of dimes to quarters is $3 : 2$. The total value of the coins is $\$2.40$. How many quarters are there?

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CHEMICAL COMPOUNDS

■ 1. Find the molar mass for one molecule of table salt in grams per mole. Table salt has the molecular formula NaCl.

Sodium (Na) has a molar mass of 22.989770 g/mol

Chlorine (Cl) has a molar mass of 35.453 g/mol

■ 2. Find the molar mass for one molecule of isopropyl chloride in grams per mole. Isopropyl chloride has the molecular formula C_3H_7Cl.

Carbon (C) has a molar mass of 12.0107 g/mol

Hydrogen (H) has a molar mass of 1.00794 g/mol

Chlorine (Cl) has a molar mass of 35.453 g/mol

■ 3. Find the mass of hydrogen in a 185 g ammonium carbonate sample, $(NH_4)_2CO_3$, if the molar mass of ammonium carbonate is 96.09 g/mol and the molar mass of hydrogen is 1.00794 g/mol.

■ 4. Find the total mass of a sample of silver phosphate, Ag_3PO_4, in grams per mole given that it contains 25.55 g of silver with the molar mass 107.8682 g/mol. The molar mass of Ag_3PO_4 is 418.58 g/mol.

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■ 5. If vitamin C has the molecular formula $C_6H_8O_6$ and the molar mass of vitamin C is 176.12 g/mol, find the mass of each element in a 325 g vitamin C sample.

Carbon (C) has a molar mass of 12.01 g/mol

Hydrogen (H) has a molar mass of 1.00794 g/mol

Oxygen (O) has a molar mass of 15.9994 g/mol

■ 6. Find the molar mass of calcium (Ca) in one mole of calcium carbonate. $CaCO_3$ has a molar mass of 100.0869 g/mol.

Carbon (C) has a mass of 12.0107 g/mol

Oxygen (O) has a mass of 15.9994 g/mol

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FRACTIONS TO DECIMALS TO PERCENTS

- 1. Convert $60\,\%$ to a fraction in lowest terms.

- 2. Convert $33.5\,\%$ to a decimal.

- 3. Convert $2/3$ to a percent.

- 4. Find $15\,\%$ of 48.

- 5. Find a mixed fraction that represents $8\,\%$ of 120.

- 6. Convert $100/160$ to a percent.

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PERCENT MARKUP

■ 1. A book store purchases a book for $\$6.00$ and sells it for $\$9.00$. What percentage of the original price is the markup amount?

■ 2. A bike shop buys a used bike for $\$130$ and marks up the price by $35\,\%$. What is the markup amount?

■ 3. It costs a car manufacturer $\$12{,}800$ to produce a car. The percent markup is $48\,\%$. What is the selling price of the car.

■ 4. A bakery purchases a dozen sugar cookies for $\$2.25$. The markup percent is $60\,\%$. What is the selling price of the dozen sugar cookies?

■ 5. A store purchases dresses from a manufacturer, marks them up by $75\,\%$, and sells each dress for $\$91$. How much did the store pay the manufacturer for each dress?

■ 6. If a furniture store purchases a chair from a manufacturer, marks it up by $24\,\%$, and sells the chair for $\$84.94$. How much did the furniture store pay the manufacturer for the chair?

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PERCENT MARKDOWN

■ 1. A computer has an original price of $\$375$ and is now on sale for $\$255$. What is the percent markdown?

■ 2. A sweater has an original price of $\$34$ and is now on sale for $\$25.50$. What is the percent markdown?

■ 3. The regular price of an item is $\$75$, but the item is now on clearance for $40\,\%$ off the regular price. What is the sale price of the item?

■ 4. The regular price of the latest smartphone is $\$749$. After two years, the smartphone is on sale for $25\,\%$ off. What is the sale price of the item?

■ 5. The price of a house was marked down to $\$250{,}000$, and the sale price was $12\,\%$ off of the original price. What was the original price of the house?

■ 6. The sale price of a shirt is $\$68.00$, and the shirt is on sale for $25\,\%$ off the original price. What was the original price of the shirt?

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CALCULATING COMMISSION

■ 1. A makeup company advertises that we can make 15% commission on sales of their product. If we sell $\$3{,}252$ worth of product, how much money do we earn?

■ 2. An employee at a clothing store earned $\$1{,}450$ in hourly pay for the month. She also sold $\$4{,}250$ worth of merchandise and will earn a commission of 6% on those sales. What is the employee's expected paycheck before tax deductions?

■ 3. A local bakery sells croissants for $\$5.00$ each. A sales clerk makes a 6% commission on the selling price of each croissant he sells. How many croissants does he need to sell to earn $\$60$ in commission?

■ 4. A car salesman earns $\$48{,}000$ per year plus a commission of 12% on all the cars he sells. If he wants a yearly salary of $\$72{,}500$, how much money in car sales does he need to make?

■ 5. Brittany earns $\$1{,}772.10$ in commission of makeup products. If she earns 18% commission, how much money in makeup sales did she make?

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6. Anthony works at a clothing store and earned $\$1{,}644.75$ last month before tax deductions. If he earns $7.5\,\%$ in commission and his hourly pay was $\$975$ for the month, how much clothing did he sell?

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CALCULATING SIMPLE INTEREST

■ 1. If we deposit $\$300$ into a savings account and it earns $2\,\%$ in simple interest, how much interest will we earn on the account in 7 years?

■ 2. If we invest $\$500$ that earns $13\,\%$ in simple interest, how much interest will we earn in 12 years?

■ 3. What is the simple interest rate if we invest $\$7{,}000$ and earn $\$3{,}250$ in interest in 15 years?

■ 4. If we deposit $\$275$ into a savings account that earns $4\,\%$ simple interest, how much is in the account after 2 years?

■ 5. If we invest $\$450$ that earns $15\,\%$ simple interest, how many years will it take to have $\$1{,}800$ in the account?

■ 6. If we invest $\$1{,}230$ that earns $14\,\%$ simple interest, how much is in the account after 10 years?

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COMPLEX FRACTIONS

■ 1. Simplify the expression.

$$\frac{\frac{3}{5}}{\frac{2}{7}}$$

■ 2. Simplify the expression.

$$\frac{y}{\frac{x}{z}}$$

■ 3. Simplify the expression.

$$\frac{\frac{x}{b}}{n}$$

■ 4. Simplify the expression.

$$\frac{\frac{a}{m}}{n+\frac{1}{b}}$$

■ 5. Simplify the expression.

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$$\frac{\frac{1}{y}-\frac{1}{x}}{1-\frac{1}{y}}$$

6. Simplify the expression.

$$\frac{\frac{1}{a-5}-\frac{1}{a+5}}{\frac{5}{a+5}}$$

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PROPORTIONS WITH COMPLEX FRACTIONS

■ 1. Solve for the variable.

$$\frac{x}{\frac{1}{2}} = \frac{\frac{3}{4}}{\frac{1}{4}}$$

■ 2. Solve for the variable.

$$\frac{\frac{4}{7}}{\frac{1}{6}} = \frac{y}{\frac{7}{2}}$$

■ 3. Solve for the variable.

$$\frac{\frac{x}{2}}{\frac{8}{3}} = \frac{\frac{3}{4}}{\frac{2}{5}}$$

■ 4. Solve for the variable.

$$\frac{\frac{3}{8}}{\frac{x}{2}} = \frac{\frac{1}{4}}{\frac{4}{5}}$$

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5. Solve for the variable.

$$\frac{\frac{4}{5}-\frac{1}{2}}{\frac{3}{2}}=\frac{\frac{6}{7}+\frac{1}{7}}{\frac{b}{8}}$$

6. Solve for the variable.

$$\frac{\frac{2}{3}}{\frac{1}{c}}=\frac{\frac{4}{5}}{\frac{7}{6}}$$

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IMAGINARY AND COMPLEX NUMBERS

1. Simplify the imaginary expression.

$$2 - 6i - 4 + 9i$$

2. Simplify the imaginary expression.

$$-3 - 7i + 8 + 3i$$

3. Simplify the imaginary expression.

$$\sqrt{-4} + ii + 5i - 2i^3$$

4. Simplify the imaginary expression.

$$\sqrt{27} - 3ii + 2i - 7i^3 + \sqrt{-36}$$

5. Simplify the imaginary expression.

$$\sqrt{-9} + 2i^3 + 6i - \sqrt{25}\sqrt{-25} - 2\sqrt{-16}$$

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6. Simplify the imaginary expression.

$$\sqrt{-4} + 2i^4 + 6i^5 - \sqrt{-49} - 2i^6$$

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RATIONALIZING COMPLEX DENOMINATORS

1. Use the conjugate method to simplify the imaginary expression.

$$\frac{2+6i}{3-i}$$

2. Use the conjugate method to simplify the imaginary expression.

$$\frac{2-2i}{4i-1}$$

3. Use the conjugate method to simplify the imaginary expression.

$$\frac{3i+2i^2}{5i^3+4i^4}$$

4. Use the conjugate method to simplify the imaginary expression.

$$\frac{2i+4i^2}{6-6i}$$

5. Use the conjugate method to simplify the imaginary expression.

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$$\frac{\sqrt{-5}\sqrt{-5}-7i^3}{3+i}$$

6. Use the conjugate method to simplify the imaginary expression.

$$\frac{\sqrt{-2}\sqrt{-2}+3i^3}{i-4}$$

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COEFFICIENTS IN QUADRATICS

■ 1. Factor the quadratic.

$$6x^2 + 11x - 10$$

■ 2. Factor the quadratic.

$$20x^2 - 23x + 6$$

■ 3. Factor the quadratic.

$$4x^2 + 26x + 36$$

■ 4. Factor the quadratic.

$$14x^2 + 15x + 4$$

■ 5. Factor the quadratic.

$$12x^2 + 4x - 1$$

■ 6. Factor the quadratic.

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$8x^2 - 10x - 63$

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GROUPING

■ 1. Factor the expression by grouping.

$$2x - 3y - 4ax + 6ay$$

■ 2. Factor the quadratic by grouping.

$$4x^2 + 2xy + 10x + 5y$$

■ 3. Factor the expression by grouping.

$$8ab + 2b - 4a - 1$$

■ 4. Factor the expression by grouping.

$$9z + 9qr + 5ayz + 5ayqr$$

■ 5. Factor the quadratic by grouping.

$$3k^2 + 7k - 6$$

■ 6. Factor the quadratic by grouping.

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$6x^2 + 13x - 5$

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DIFFERENCE OF CUBES

■ 1. Factor the polynomial.

$$x^3 - 27y^9$$

■ 2. Factor the polynomial.

$$8x^3y^6 - 64z^{21}$$

■ 3. Factor the polynomial.

$$a^3b^{12} - 125c^6$$

■ 4. Factor the polynomial.

$$27y^6z^3 - 216x^9$$

■ 5. Factor the polynomial.

$$8x^{15} - 27y^9$$

■ 6. Factor the polynomial.

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$216a^3b^6 - 125c^{24}d^3$

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SUM OF CUBES

■ 1. If $x^2 - 2xy^2 + 4y^4 = 5$ and $x + 2y^2 = 8$, what is the value of $8x^3 + 64y^6$?

■ 2. Factor the polynomial.

$$216a^{21} + 64b^{15}c^9$$

■ 3. Factor the polynomial.

$$512z^{24} + 125m^6r^3$$

■ 4. Factor the polynomial.

$$64j^3k^6 + 8r^{12}t^6$$

■ 5. Factor the polynomial.

$$729x^{18} + 216y^6$$

■ 6. Factor the polynomial.

$$(x - 5)^3 + 125$$

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SIMPLIFYING RATIONAL FUNCTIONS

■ 1. Simplify the rational function to lowest terms.

$$\frac{x^4 + x^3 - 6x^2}{x^3 + x^2 - 2x}$$

■ 2. Reduce the fraction to its lowest terms.

$$\frac{10x^2 - 5x + 20}{15x^2}$$

■ 3. Reduce the fraction to its lowest terms.

$$\frac{18y^2 + 6y}{8y}$$

■ 4. Simplify each expression in the difference.

$$\frac{3ab + 2a^2b^2}{5ab} - \frac{12a^3b^3 + 3a^2b^2}{6a^2b^2}$$

■ 5. Simplify each expression in the sum.

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$$\frac{2ab^2+3a^2b^3}{a^3b^3}+\frac{2ab^3+b^4}{ab^3}$$

6. Simplify each expression in the difference.

$$\frac{21x^2y^2}{14x^3y}-\frac{24xy+12y}{96y}$$

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ADDING AND SUBTRACTING RATIONAL FUNCTIONS

■ 1. Simplify the expression.

$$\frac{2}{3ab}+\frac{b}{4a}+\frac{ab}{6}$$

■ 2. Simplify the expression.

$$\frac{x+1}{x-1}+\frac{2x}{x-5}+\frac{x+2}{x^2-6x+5}$$

■ 3. Simplify the expression.

$$\frac{a}{3xy}+\frac{b}{15y^2}+\frac{c}{5x^3y^2}$$

■ 4. Simplify the expression.

$$\frac{x}{2x^2y}+\frac{y}{3z}+\frac{z}{5yz^2}$$

■ 5. Simplify the expression.

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$$\frac{3ab}{4c}+\frac{2bc}{6a^3}+\frac{5}{8ab^2c^3}$$

6. Simplify the expression.

$$\frac{x}{x+6}+\frac{x-6}{x}$$

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FACTORING TO FIND A COMMON DENOMINATOR

■ 1. Simplify the expression by combining the two fractions.

$$\frac{x+1}{2x^2+5x-3}+\frac{2}{x+3}$$

■ 2. What is the common denominator of the rational expressions?

$$\frac{x^2-1}{x^2-4} \text{ and } \frac{x+1}{3x^2-3x-6}$$

■ 3. Simplify the expression by combining the two fractions.

$$\frac{3}{x-2}-\frac{x-4}{x^2-5x+6}$$

■ 4. Fill in the blank with the correct expression.

$$\frac{2}{_____}-\frac{x-2}{x^2-9}=\frac{2(x-3)-4(x-2)}{4(x-3)(x+3)}$$

■ 5. Simplify the expression by combining the two fractions.

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$$\frac{4}{x^2-2x-3}-\frac{1}{x^2+5x+4}$$

■ 6. What went wrong in the following simplification?

$$\frac{3}{x^2-25}-\frac{1}{x+5}$$

$$\frac{3-x-5}{(x-5)(x+5)}$$

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MULTIPLYING RATIONAL FUNCTIONS

■ 1. Simplify the expression.

$$\frac{25x^2-4}{x^2-36}\cdot\frac{x+6}{5x-2}$$

■ 2. Simplify the expression.

$$\frac{4x^2-49}{9x^2-16}\cdot\frac{3x+4}{2x+7}$$

■ 3. Simplify the expression.

$$\frac{x^2+8x+16}{9x^2+36x+36}\cdot\frac{3x+6}{x+4}$$

■ 4. Simplify the expression.

$$\frac{16x^2+16x+4}{x^2+18x+81}\cdot\frac{x^2-81}{16x^2-4}$$

■ 5. Simplify the expression.

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$$\frac{x^2+5x-14}{x^2+2x-3}\cdot\frac{x^2+4x-5}{x^2+9x+14}$$

6. Simplify the expression.

$$\frac{2x^2-13x-24}{3x^2-x-4}\cdot\frac{3x^2-7x+4}{x^2-6x-16}$$

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DIVIDING RATIONAL FUNCTIONS

■ 1. Simplify the expression.

$$\frac{2x+16}{9x^2+27x} \div \frac{3x+24}{x+3}$$

■ 2. Simplify the expression.

$$\frac{3x^3-3x^2-6x}{2x^2-14x+24} \div \frac{3x^2+21x}{x^2-8x+15}$$

■ 3. Simplify the expression.

$$\frac{2x^2-13x-7}{12x+6} \div \frac{3x-2}{3x^2-17x+10}$$

■ 4. Simplify the expression.

$$\frac{4x^2+13x+10}{3x+6} \div \frac{3x-1}{3x^2-13x+4}$$

■ 5. Simplify the expression.

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$$\frac{4x^2-9}{x^2+12x+36} \div \frac{4x^2-12x+9}{x^2+7x+6}$$

6. Simplify the expression.

$$\frac{15x^2+75x+90}{5x^2+50x+125} \div \frac{x^2-3x+2}{x^2-25}$$

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DIRECT VARIATION

■ 1. If $10k = 5$ and $kx = 3$, find x.

■ 2. If x and y vary directly and the constant of variation, k, equals $1/3$, what is the value of y when $x = 54$, assuming the direct variation relationship between the variables is given by $y = kx$?

■ 3. A restaurant takes in $\$15{,}000$ in a 5 hour period. Write a direct variation equation for the relationship between income and number of hours. Estimate how many hours it would take the restaurant to earn $\$35{,}750$.

■ 4. If x varies directly with y and $y = 4$ when $x = 20$, what is the value of the constant of variation, k, assuming the direct variation relationship between the variables is given by $y = kx$?

■ 5. If x varies directly with y and $y = 15$ when $x = 5$, what is the value of x when $y = 36$, assuming the direct variation relationship between the variables is given by $y = kx$?

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6. If x varies directly with y and $y = 7$ when $x = 42$, what is the value of y when $x = 54$, assuming the direct variation relationship between the variables is given by $y = kx$?

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ADDING AND SUBTRACTING RATIONAL FUNCTIONS

■ 1. Simplify the expression.

$$\frac{2}{3ab}+\frac{b}{4a}+\frac{ab}{6}$$

■ 2. Simplify the expression.

$$\frac{x+1}{x-1}+\frac{2x}{x-5}+\frac{x+2}{x^2-6x+5}$$

■ 3. Simplify the expression.

$$\frac{a}{3xy}+\frac{b}{15y^2}+\frac{c}{5x^3y^2}$$

■ 4. Simplify the expression.

$$\frac{x}{2x^2y}+\frac{y}{3z}+\frac{z}{5yz^2}$$

■ 5. Simplify the expression.

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$$\frac{3ab}{4c}+\frac{2bc}{6a^3}+\frac{5}{8ab^2c^3}$$

6. Simplify the expression.

$$\frac{x}{x+6}+\frac{x-6}{x}$$

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INVERSE VARIATION

■ 1. If $k/3 = 6$ and $k/x = 2$, find x.

■ 2. The length of the base of a triangle with constant area varies inversely with the triangle's height. When the base is 5 cm long, the height is 6 cm. Find the length of the base when the height is 3 cm.

■ 3. If x and y vary inversely and the constant of variation, k, equals $1/3$, what is the value of y when $x = 8$?

■ 4. If x varies inversely with y and $y = 5$ when $x = 6$, what is the value of the constant of variation, k?

■ 5. If x varies inversely with y and $y = 4$ when $x = 2$, what is the value of x when $y = 1/2$?

■ 6. If x varies inversely with y and $y = 3$ when $x = 9$, what is the value of y when $x = 1/4$?

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DECIMAL EQUATIONS

■ 1. Solve the decimal equation.

$$0.34x - 0.62 = 1.25$$

■ 2. Solve the decimal equation.

$$0.1(2.1a - 1.4a) + 3.57 = 2.8$$

■ 3. Solve the decimal equation.

$$4a + 6a = 1.7$$

■ 4. Solve the decimal equation.

$$0.12n + 3.6 = 4.8$$

■ 5. Solve the decimal equation.

$$5n - 6.1 = -2.9$$

■ 6. Solve the decimal equation.

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$$3.2x + 2.6 = 1.8x - 4.4$$

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FRACTIONAL EQUATIONS

■ 1. Solve for the variable.

$$2x - 5 = \frac{4x + 3}{5}$$

■ 2. Solve for the variable.

$$\frac{4}{3}x = 18$$

■ 3. Solve for the variable.

$$\frac{3}{4}x + \frac{5}{4} = \frac{7}{8}$$

■ 4. Solve for the variable.

$$\frac{4}{7}x + \frac{1}{7} = \frac{10}{7}$$

■ 5. Solve for the variable.

$$\frac{1}{2}a - \frac{5}{4}a = -\frac{10}{3} + \frac{5}{2}a$$

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6. Solve for the variable.

$$\frac{1}{2}\left(\frac{1}{2}x-\frac{1}{3}\right)=\frac{7}{3}+\frac{9}{2}$$

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RATIONAL EQUATIONS

■ 1. Solve the equation.

$$\frac{x-3}{x+3}=\frac{4}{5}$$

■ 2. Solve the equation.

$$\frac{x}{6}-\frac{5}{3x}=\frac{1}{4}$$

■ 3. Solve the equation.

$$\frac{2}{x+5}=\frac{4}{x-1}$$

■ 4. Solve the equation.

$$2+\frac{6}{x-7}=-\frac{8}{x+3}$$

■ 5. Solve the equation.

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$$\frac{5}{a-4}-\frac{3}{a+4}=-\frac{1}{a^2-16}$$

6. Solve the equation.

$$\frac{1}{2x^2}+\frac{3}{4x}=\frac{x+7}{x^2}$$

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RADICAL EQUATIONS

■ 1. Solve the radical equation for the variable.

$$\sqrt[3]{2x-1}+5=7$$

■ 2. Solve the radical equation for the variable.

$$2\sqrt{x}=14$$

■ 3. Solve the radical equation for the variable.

$$\sqrt{x+1}-3=2$$

■ 4. Solve the radical equation for the variable.

$$\sqrt{x-3}+2=\sqrt{2x+1}$$

■ 5. Solve the radical equation for the variable.

$$\sqrt{1-x}-x=5$$

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6. Solve the radical equation for the variable.

$$\sqrt{x^2 - 2x + 4} + 4 = x$$

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MULTIVARIABLE EQUATIONS

1. Solve for x if $y = z/x$.

2. Solve for t if $4s - 3t + u = 5$.

3. Solve for y if $z - x + 4y = 3x + z$.

4. Solve for c if $2a - b + 3c = 2b - 4a + c$.

5. Solve for y if $2x - y + z = 3x$.

6. Solve for a if $x + y = 3ab + c$.

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MULTIVARIABLE RATIONAL EQUATIONS

■ 1. Solve the abstract equation for x, if $x \neq 0$.

$$\frac{1}{x} - z = y$$

■ 2. Solve the abstract equation for y, if $x \neq 0$.

$$\frac{y}{x} + 3x = 2z$$

■ 3. Solve the abstract equation for a, if $a \neq 0$ and $b \neq 0$.

$$\frac{bc}{a} - cxy = \frac{z}{b}$$

■ 4. Solve the abstract equation for y, if $y \neq 0$, $b \neq 0$, and $n \neq 0$.

$$\frac{1}{y} + \frac{a}{b} = \frac{m}{n}$$

■ 5. Solve the abstract equation for x, if $z \neq 0$, $n \neq 0$, and $b \neq 0$.

$$\frac{2x + y}{z} - \frac{m}{n} = \frac{a}{b}$$

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6. Solve the abstract equation for x, if $x \neq 0$ and $y + z \neq 0$.

$$\frac{1}{x} + \frac{2}{y+z} = 3$$

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SYSTEMS WITH SUBSCRIPTS

■ 1. Solve the system of equations using any method.

$$d_t = x_t - 3$$

$$d_t = \frac{1}{2}x_t + 2$$

■ 2. Solve the system of equations using any method.

$$4A_x - B_y = 15$$

$$B_y = 12 - 5A_x$$

■ 3. Solve the system of equations using any method.

$$2\,T_x + 5\,T_y = 120.8$$

$$T_y = 17\,T_x + 17.2$$

■ 4. Solve the system of equations using any method.

$$T_u R_u = 480$$

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$$R_d = \frac{1}{2}R_u$$

$$T_d R_d = 320$$

$$T_d = 2 + T_u$$

■ 5. Solve the system of equations using any method.

$$2X_1 + 3Y_1 = 1$$

$$X_2 = 4X_1$$

$$X_2 + Y_2 = 9$$

$$Y_2 = Y_1 + 2$$

■ 6. Solve the system of equations using any method.

$$6X_1 + 12Y_1 = 78$$

$$X_2 = 2X_1$$

$$2X_2 - 3Y_2 = -24$$

$$Y_2 = 3Y_1 - 3$$

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UNIFORM MOTION PROBLEMS

■ 1. Kaitlyn is driving at a constant rate of 55 mph on the highway. 1 hour later, her friend Charlotte starts from the same point and drives at a constant rate of 65 mph. How many hours will each woman need to travel before Charlotte catches Kaitlyn. At that point, how far have each of them traveled?

■ 2. Sam walked 4 mph from the bus stop to the school. John walked 5 mph from the same bus stop to the school and it took him 1 hour to get there. How long did it take Sam to get to the school?

■ 3. McKenzie and Daisy plan a trip to the mountains together. McKenzie travels at 45 mph and Daisy travels at 60 mph. Daisy's trip took 3 hours less than McKenzie's. How far did each of them travel?

■ 4. Talan and Emily participate in a race together. Talan runs 10 mph and finishes 1 hour and 18 minutes before Emily does. If Emily runs 5 mph, how long does it take each person to finish the race and how far did they run?

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■ 5. Adeline and Ellie live 10 miles away from each other. Adeline started walking towards Ellie at $1:00$ p.m.. Ellie left 1 hour later and walked at a rate of 4 mph. If they met at $3:00$ p.m., how fast did Adeline walk?

■ 6. A train traveled 420 miles at 48 mph and arrived 1 hour and 45 minutes later than it was scheduled to arrive. How fast should the train have traveled in order to arrive on time?

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NUMBER WORD PROBLEMS

■ 1. The sum of two consecutive odd integers is -8. What are the two numbers?

■ 2. The product of two consecutive integers is 182. What are the two numbers if they are both negative?

■ 3. The sum of the digits of a two-digit counting number is 11. When the digits are reversed, the new number is 63 greater than the original number. What was the original number?

■ 4. The sum of the digits of a two-digit counting number is 14. The ratio of the units digit to the tens digit is 3 to 4. What is the number?

■ 5. The sum of the digits of a two-digit counting number is 8. The ratio of the tens digit to the units digit is 3 to 1. What is the number?

■ 6. Find three negative consecutive odd integers such that the sum of the first and third is 49 less than the product of the second and third.

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AGE WORD PROBLEMS

■ 1. In 16 years, Thorin will be nine times older than he is now. How old is he now?

■ 2. In 12 years, Jake will be four times as old as he was nine years ago. How old will he be in 12 years?

■ 3. In 16 years, Erica will be six times as old as she was 14 years ago. How old was she four years ago?

■ 4. 10 years ago, Tim was three years younger than twice Sally's age. 17 years from now Sally will be seven years older than 5/8 of Tim's age. How old are Sally and Tim now?

■ 5. Jessica is five years younger than Ryan. 12 years ago, Jessica was half the age Ryan will be in three years. How old is Jessica now?

■ 6. Kate is five times older than Sam. Five years ago, Kate was 15 times older than Sam. How old will Sam be in three years?

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SYSTEMS WITH NON-LINEAR EQUATIONS

■ 1. Solve the system of equations.

$$y = x - 1$$

$$x^2 + y^2 = 1$$

■ 2. Solve the system of equations.

$$y - 2x = 0$$

$$(x - 2)^2 + (y - 1)^2 = 9$$

■ 3. Solve the system of equations.

$$2y - x = -2$$

$$(x - 6)^2 + 4(y - 4)^2 = 16$$

■ 4. Solve the system of equations.

$$y = \frac{1}{2}x + 1$$

$$x^2 - y^2 = 4$$

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■ 5. Solve the system of equations.

$$y = -\frac{1}{2}x - 3$$

$$\frac{(x-3)^2}{9} - \frac{(y+3)^2}{9} = 1$$

■ 6. Solve the system of equations.

$$2y - 3x = 14$$

$$\frac{(y+2)^2}{9} + \frac{(x+4)^2}{4} = 1$$

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SYSTEMS OF THREE EQUATIONS

■ 1. Find the unique solution to the system of equations.

$$2x + y - z = 3$$

$$x - y + z = 0$$

$$x - 2y - 3z = 4$$

■ 2. Find the unique solution to the system of equations.

$$3x + y - z = -2$$

$$x - 2y + 3z = 23$$

$$2x + 3y + 2z = 5$$

■ 3. Find the unique solution to the system of equations.

$$-2x + 3y - 4z = 10$$

$$4x + 3y + 2z = 4$$

$$x - 6y + 4z = -19$$

■ 4. Find the unique solution to the system of equations.

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$$2x - y + z = 9$$

$$4x - 2y + 2z = 18$$

$$-2x + y - z = -9$$

■ 5. Find the unique solution to the system of equations.

$$x + 2y - z = 9$$

$$3x + y - z = 5$$

$$-x - 4y + z = 2$$

■ 6. Find the unique solution to the system of equations.

$$-x + y - z = 12$$

$$x - y + z = 2$$

$$2x - 2y + 2z = 9$$

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PARALLEL AND PERPENDICULAR LINES

■ 1. Determine if the lines are parallel, perpendicular, or neither.

$$y = 2x + 5$$

$$2y + x = -3$$

■ 2. Determine if the lines are parallel, perpendicular, or neither.

$$y = 5x + 1$$

$$10y - 20 = 15x$$

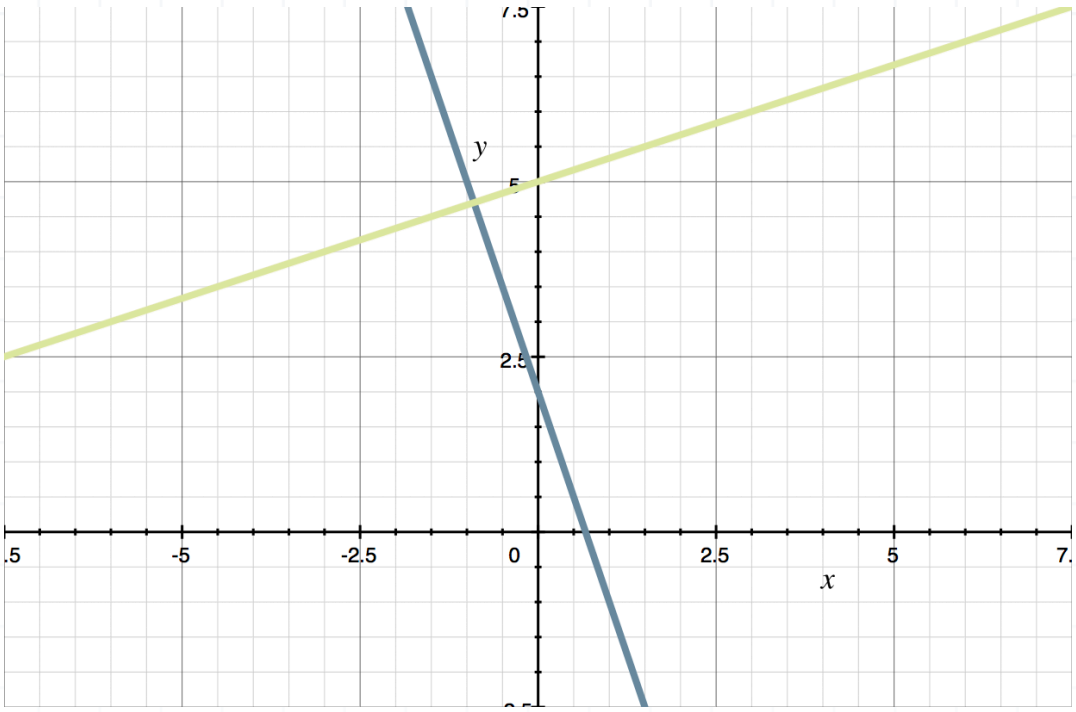
■ 3. Write the equation of the line with a y-intercept of -3 that's parallel to $7x + 3y = 12$.

■ 4. Write the equation of the line passing through $(3,6)$ and perpendicular to $6x + 2y = 4$.

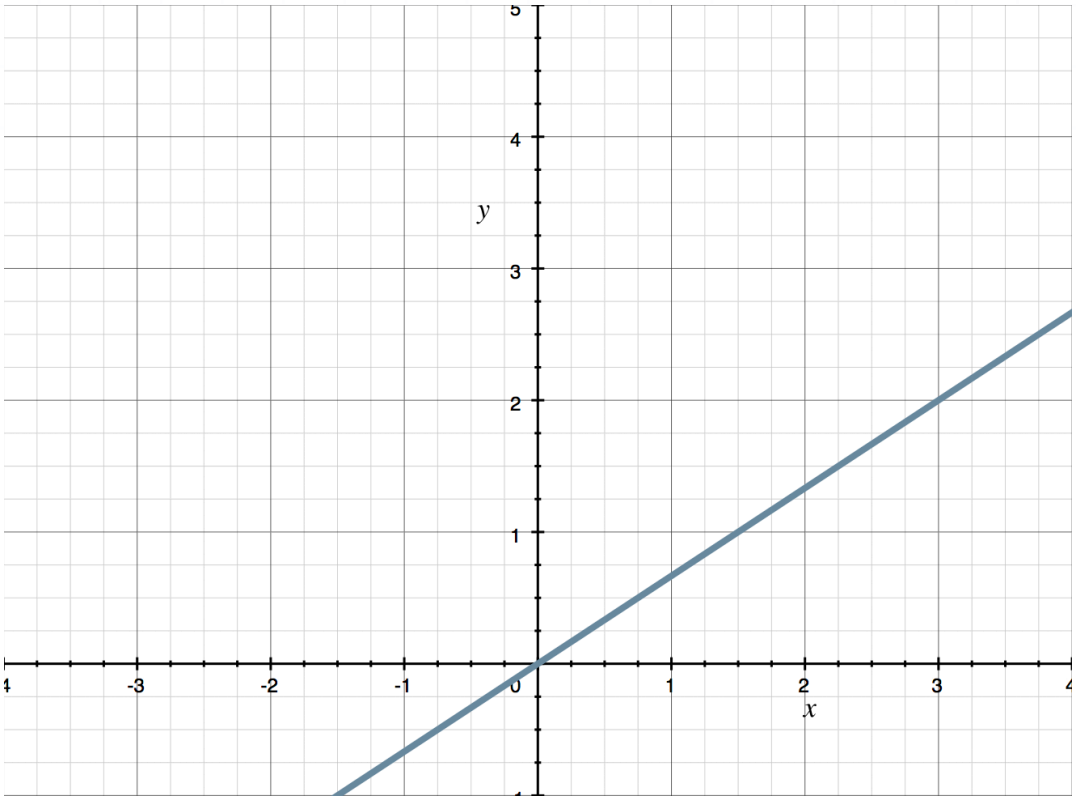
■ 5. Determine if the lines in the graph are parallel, perpendicular, or neither.

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■ 6. Graph the line with a y-intercept of 3 that's parallel to the line in the graph.



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GRAPHING PARABOLAS

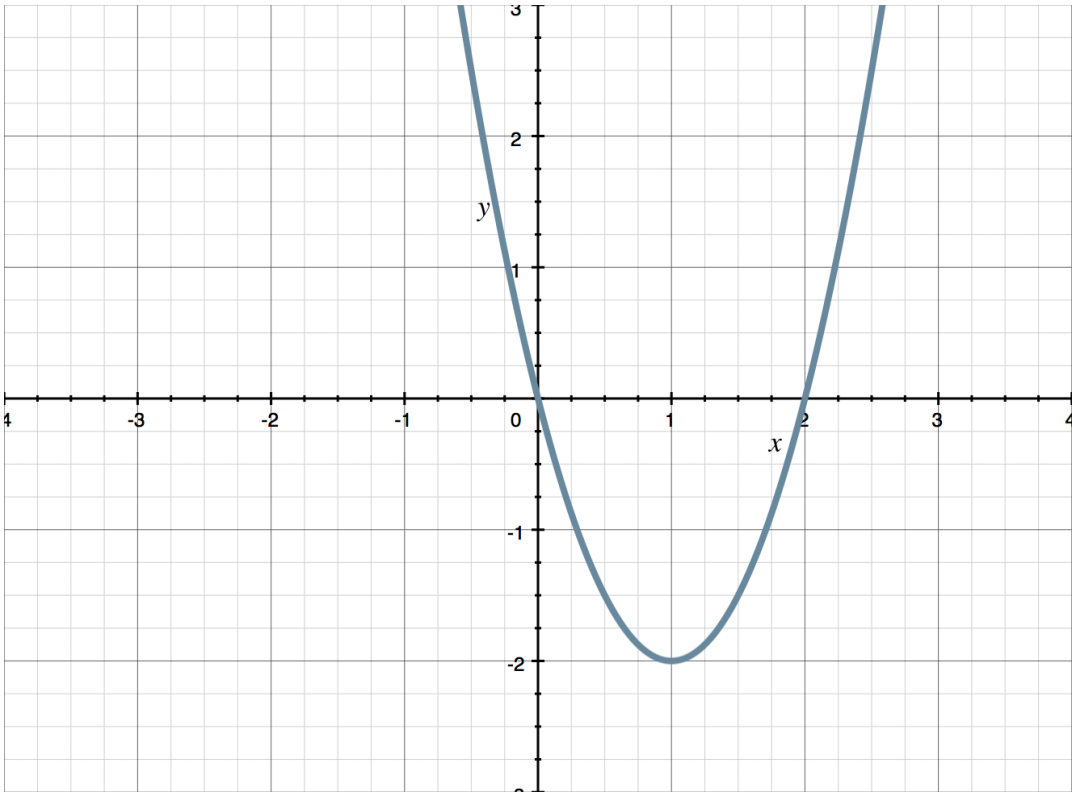
■ 1. Write the equation in vertex form.

$$y = -2x^2 + 24x - 68$$

■ 2. Find the vertex and axis of symmetry of $y = x^2 + 5x + 6$.

■ 3. Find the vertex and axis of symmetry of $y = 3(x + 2)^2 + 6$.

■ 4. Using the graph, find the equation of the parabola in standard form.



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5. Complete the square to graph the parabola $y = x^2 + 6x + 5$.

6. Complete the square to graph $y = -x^2 - 4x - 6$.

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CENTER AND RADIUS OF A CIRCLE

1. Find the center and radius of the circle.

$$x^2 + y^2 + 14x + 22y + 145 = 0$$

2. Find the center and radius of the circle.

$$16x^2 + 16y^2 - 8x - 24y - 150 = 0$$

3. Find the center and radius of the circle.

$$9x^2 + 9y^2 - 30x - 6y - 118 = 0$$

4. Find the center and radius of the circle.

$$x^2 + y^2 + 4x - 2y = 0$$

5. Find the center and radius of the circle.

$$x^2 + y^2 - 12x + 10y - 3 = 0$$

6. Find the center and radius of the circle.

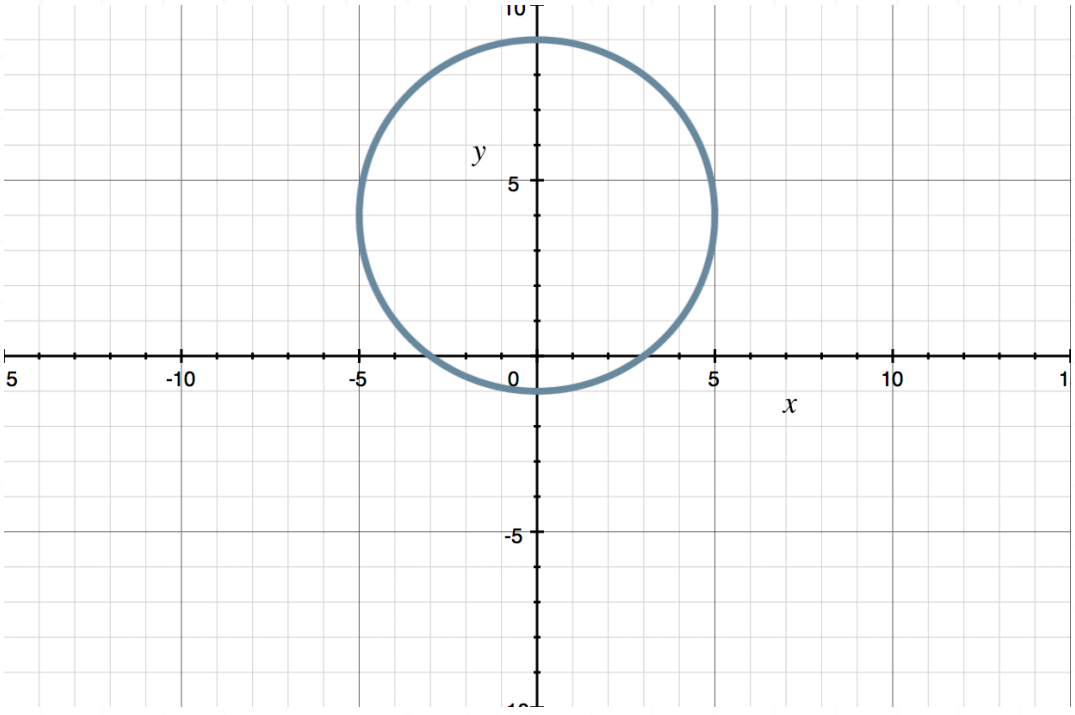
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$$16x^2 + 16y^2 + 96x - 160y + 543 = 0$$

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GRAPHING CIRCLES

■ 1. Find the equation of the circle.



■ 2. Graph the circle $(x-1)^2+(y-2)^2=4$.

■ 3. Graph the circle $x^2+(y-3)^2=16$.

■ 4. Graph the circle $x^2+y^2+2x+2y-14=0$.

■ 5. Graph the circle $x^2+y^2-8x-4y+11=0$.

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6. Graph the circle $x^2 + y^2 + 6x - 8y - 11 = 0$.

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DISTANCE BETWEEN TWO POINTS

1. What is the distance between $(-1,2)$ and $(2,6)$?

2. What is the distance between $(-7,-6)$ and $(2,3)$?

3. What is the distance between $(-1,3)$ and $(-4,8)$?

4. What is the distance between $(-3,\sqrt{7})$ and $(4,-\sqrt{7})$?

5. What is the distance between $(-\sqrt{5},\sqrt{11})$ and $(\sqrt{5},-\sqrt{11})$?

6. What is the distance between $(1,-1)$ and $(-3,3)$?

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EQUATION MODELING

■ 1. A car and a truck were driven for a week. The car traveled 75 miles more than the truck. Each vehicle had different fuel mileage. Write an equation using t (where t is the number of miles the truck traveled) to calculate the number of gallons g, used during the week.

	Car	Truck
Mileage	28 mpg	14 mpg
Distance	c miles	t miles

■ 2. A baseball is thrown at a speed of 21 ft/s straight down from a high platform. The distance it travels can be calculated using $D = 16t^2 + 21t$, where t is the amount of time in seconds that it's been falling. The average speed of any object can be calculated using $V = D/t$. Write an equation giving the time of the fall in terms of V.

■ 3. A rock is thrown at a speed of 8 ft/s straight down from a high platform. The distance it travels can be calculated using $D = 16t^2 + 8t$, where t is the amount of time in seconds that it's been falling. The average speed of any object can be calculated using $V = D/t$. Write an equation giving the time of the fall in terms of V.

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■ 4. Managers at a company are each paid $\$45{,}000$ in base salary. The company's owner wants to divide $\$162{,}000$ in annual bonus money evenly among the managers. Write an expression, in terms of the number of mangers m, that gives the amount a each manager earns per month.

■ 5. Managers at a company are each paid $\$37{,}800$ in base salary. The company's owner wants to divide $\$102{,}000$ in annual bonus money evenly among the managers. Write an expression, in terms of the number of mangers m, that gives the amount a each manager earns per month.

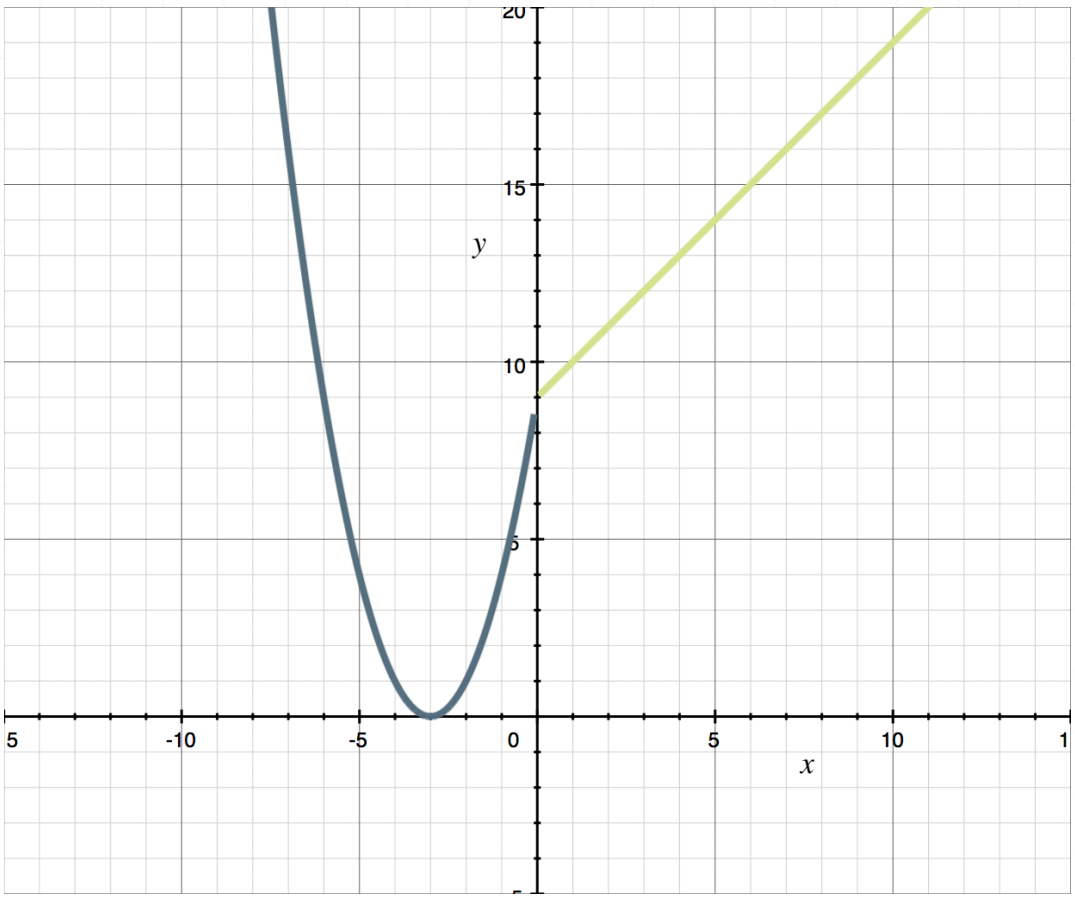
■ 6. The Jones and Anderson family go on vacation together with each family driving in their own car. The Anderson family travels 50 miles further than the Jones family. Each family averages 65 mph on the trip. Write an equation using D_a (where D_a is the total miles the Anderson family drove) to calculate the total time T both families spent driving to their destination.

	Jones	Anderson
Distance	D_j miles	D_a miles
Rate	65 mph	65 mph
Time	T_j hours	T_a hours

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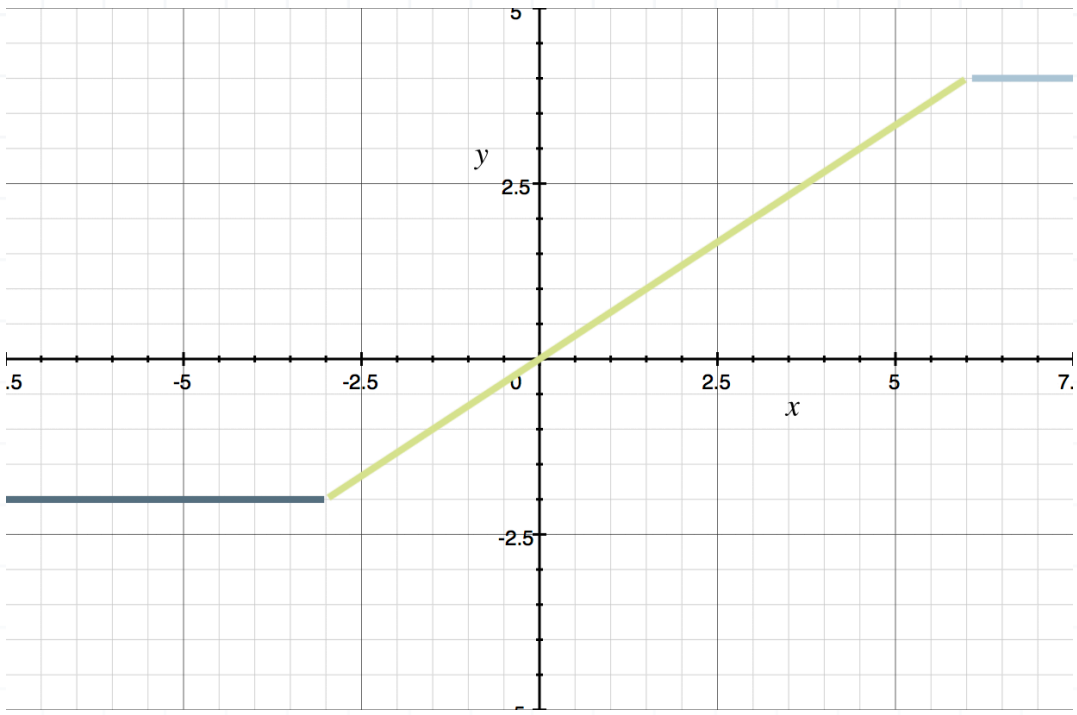
MODELING A PIECEWISE-DEFINED FUNCTION

1. Find the equation of the piecewise function.

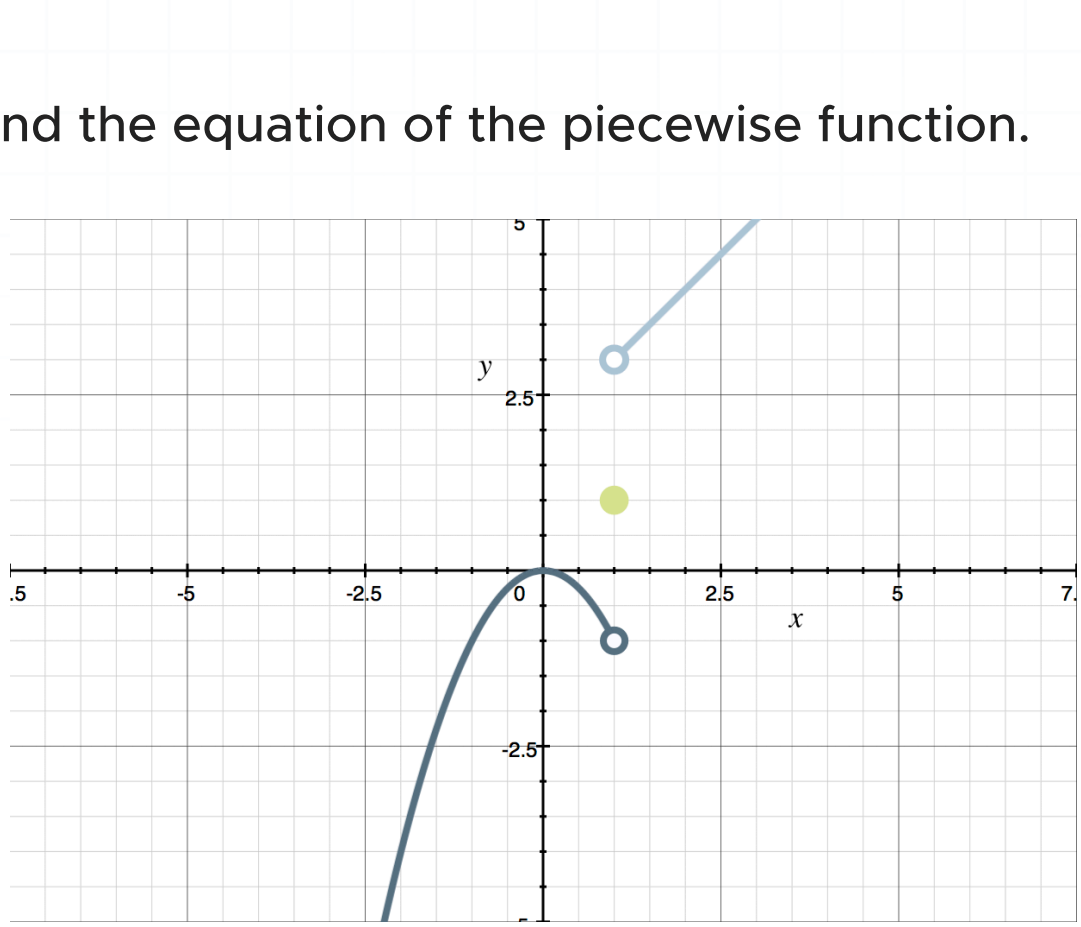


2. Find the equation of the piecewise function.

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■ 3. Find the equation of the piecewise function.



■ 4. Graph the piecewise function.

$$f(x) = \begin{cases} x^2 & x < -1 \\ \frac{1}{2}x - 3 & x \geq -1 \end{cases}$$

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5. Graph the piecewise function.

$$f(x) = \begin{cases} x^2 - 2 & x < 0 \\ 4 & x = 0 \\ -x^2 + 8 & x > 0 \end{cases}$$

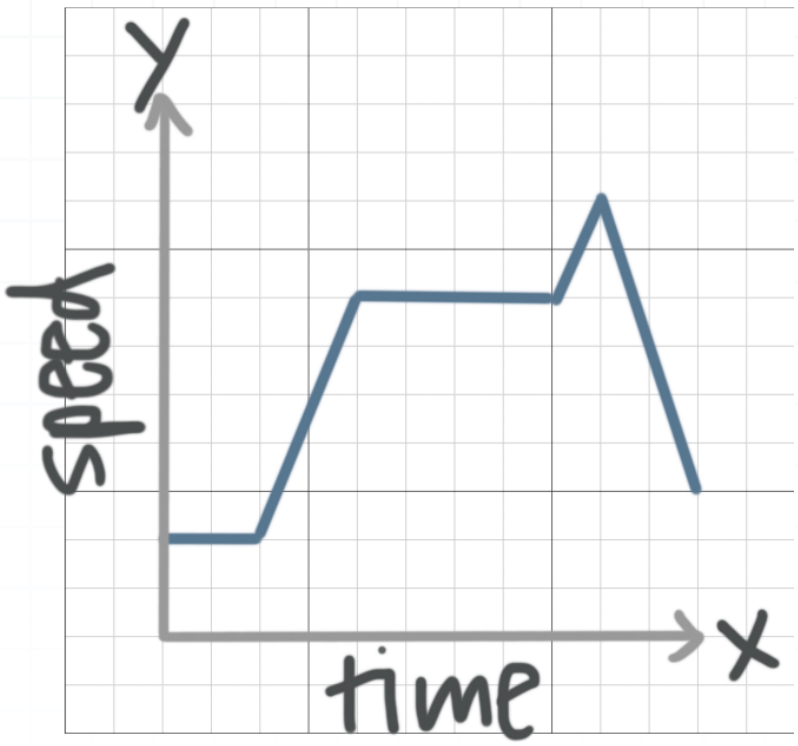
6. Graph the piecewise function.

$$f(x) = \begin{cases} -x + 1 & x < -4 \\ 5 & -4 < x \leq 3 \\ -2x + 11 & x > 3 \end{cases}$$

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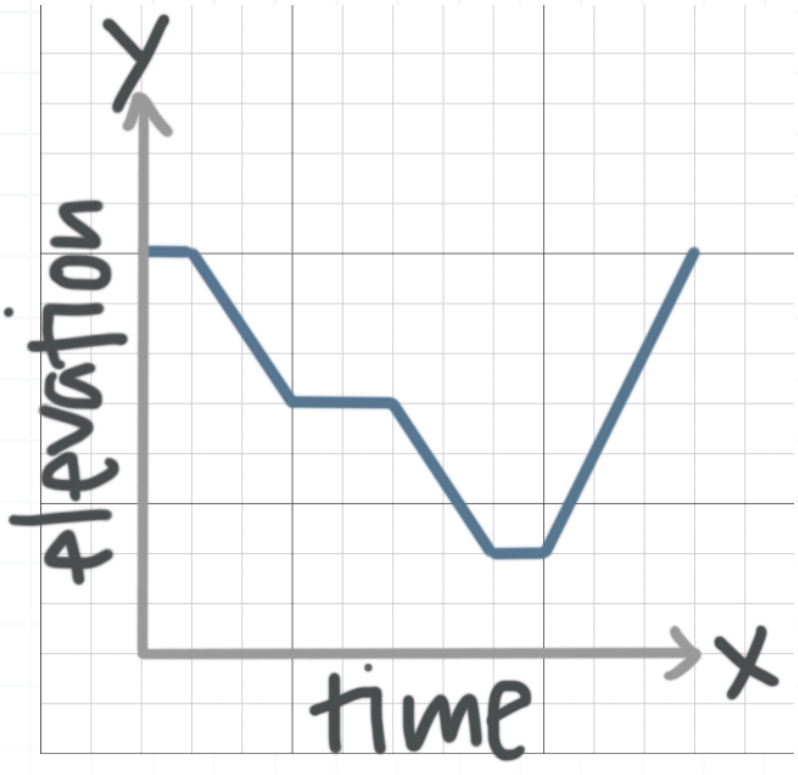
SKETCHING GRAPHS FROM STORY PROBLEMS

■ 1. A horse is practicing for a race. The graph shows the horse's speed over time. Write a possible story to go with the graph.

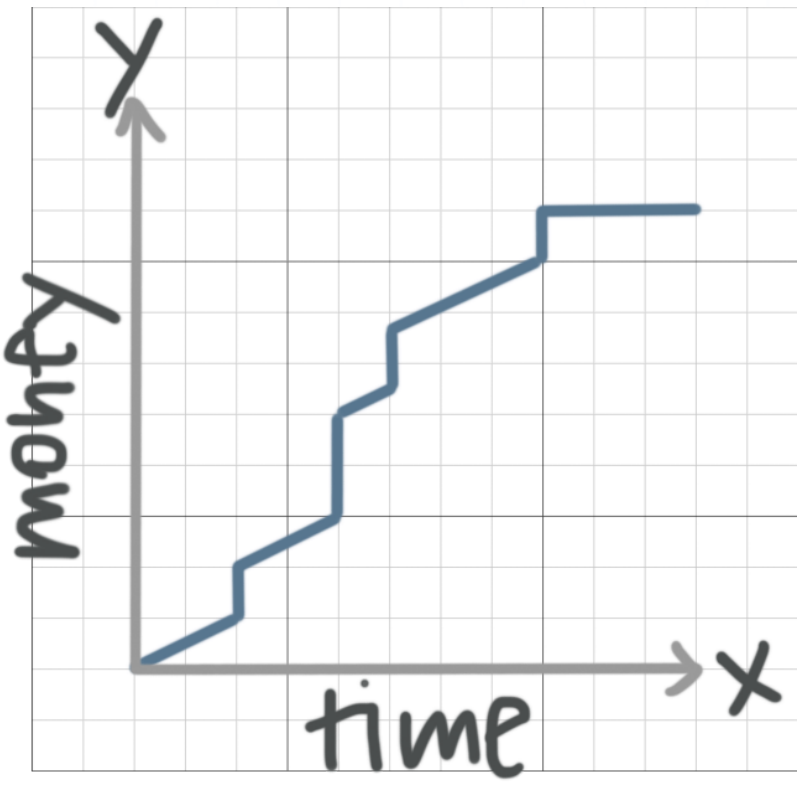


■ 2. A scuba diver takes a dive to explore the ocean. The graph below shows the diver's elevation over time. Write a possible story to go with the graph.

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■ 3. Janet delivers packages and get paid an hourly rate in addition to $1 for every package she delivers. The graph shows Janet's pay over the course of the day. Write a possible story to go with the graph.



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■ 4. A plane takes off and then cruises at $30{,}000$ feet for several hours before rising in elevation to $35{,}000$ feet to avoid turbulence for the last few hours. The plane then reaches its destination and lands. Sketch a graph representing the situation.

■ 5. The temperature throughout a summer day starts at $65°$ F at 6:00 a.m.. Over the next few hours the temperature rises steadily until it reaches $85°$ F at 1:00 p.m.. At 1:15 p.m., a rainstorm begins and cools the temperature down to $75°$ F. The temperature then steadily decreases until it reaches $70°$ F at 9:00 p.m.. Sketch a graph representing the situation.

■ 6. Brett goes for a hike up a mountain. He starts hiking up steadily for several hours with two stops for water. Then Brett stops for an hour to eat lunch and rest. He then continues up the mountain, summits, and spends a little time at the top of the mountain before climbing down. Sketch a graph representing Brett's elevation over time.

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QUADRATIC INEQUALITIES

1. Solve $x^2 - 6x + 9 \leq 0$ graphically.

2. Solve $-x^2 + 4 > 0$ graphically.

3. Solve $x^2 - 4x + 6 \geq 0$ graphically.

4. Solve $5x - x^2 \leq 0$ algebraically.

5. Solve $x^2 - x - 24 > x$ algebraically.

6. Solve $-x^2 - 3x + 6 \geq 0$ algebraically.

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SYSTEMS WITH QUADRATIC INEQUALITIES

■ 1. Sketch the solution to the system of inequalities.

$$y < x + 3$$

$$y > x^2 - 2x + 1$$

■ 2. Sketch the solution to the system of inequalities.

$$y \geq x^2 - 4x$$

$$x < 2$$

■ 3. Sketch the solution to the system of inequalities.

$$4y \leq x + 8$$

$$x^2 + 4x + 4y > 0$$

■ 4. Sketch the solution to the system of inequalities.

$$y \geq 0$$

$$y \geq x^2 + 2x + 2$$

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■ 5. Sketch the solution to the system of inequalities.

$$x - 3y < -9$$

$$y \leq (x-2)^2$$

■ 6. Sketch the solution to the system of inequalities.

$$y < 2x + 1$$

$$y > 2x + 1 - x^2$$

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COMBINATIONS OF FUNCTIONS

■ 1. Find $(f+g)(x)$.

$$f(x) = 2x^2 - x + 5$$

$$g(x) = x^2 + 4x - 7$$

■ 2. Find $(f-g)(2)$.

$$f(x) = 4x^2 - 2$$

$$g(x) = 3x^2 - 5x$$

■ 3. Find $(f-g)(x)$.

$$f(x) = x^2 - 3x + 1$$

$$g(x) = 2x - 3$$

■ 4. Find $(f \cdot g)(x)$.

$$f(x) = 2x - 3$$

$$g(x) = 3x^2 + 2$$

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■ 5. Find $(f \div g)(x)$.

$$f(x) = x^2 + 6x$$

$$g(x) = x$$

■ 6. Find $(g \div f)(x)$.

$$f(x) = x^2 + 6x$$

$$g(x) = x$$

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COMPOSITE FUNCTIONS

■ 1. Find the composite function $(g \circ f)(x)$.

$$f(x) = \sqrt{2x - 1}$$

$$g(x) = 3x^2$$

■ 2. Find $f(g(x)) - g(f(x))$.

$$f(x) = x^2 - 4x + 3$$

$$g(x) = 2x + 1$$

■ 3. Find the composite function $(g \circ h)(x) - (h \circ h)(x)$.

$$g(x) = \frac{8}{x^3}$$

$$h(x) = \sqrt[3]{x + 4}$$

■ 4. Find the composite function $(h \circ g)(x)$.

$$g(x) = \frac{8}{x^3}$$

$$h(x) = \sqrt[3]{x + 4}$$

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■ 5. Find the composite function $g(g(x))$.

$$g(x) = \frac{1}{x}$$

$$h(x) = 3x^2 - x$$

■ 6. Find the composite functions $h(g(2))$ and $g(h(2))$.

$$g(x) = \frac{1}{x}$$

$$h(x) = 3x^2 - x$$

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DOMAINS OF COMPOSITE FUNCTIONS

■ 1. What is the domain of $f \circ g$?

$$f(x) = \frac{1}{x}$$

$$g(x) = x + 5$$

■ 2. What is the domain of $f \circ g$?

$$f(x) = \frac{2}{x - 1}$$

$$g(x) = \sqrt{x - 4}$$

■ 3. What is the domain of $f \circ g$?

$$f(x) = \frac{1}{x} + 4$$

$$g(x) = \frac{3}{2x - 7}$$

■ 4. What is the domain of $f \circ g$?

$$f(x) = \frac{2}{x - 3}$$

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$$g(x) = \frac{4}{x+2}$$

■ 5. What is the domain of $f \circ g$?

$$f(x) = \frac{1}{x^2 - 3}$$

$$g(x) = \sqrt{x-1}$$

■ 6. What is the domain of $f \circ g$?

$$f(x) = 2x^2 - x + 1$$

$$g(x) = x - 3$$

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DECOMPOSING COMPOSITE FUNCTIONS

■ 1. Write $f(x)$ as the composition of two functions.

$$f(x) = \frac{4}{(2x^2 - 5x)^3}$$

■ 2. Write $f(x) = \ln(\ln x)$ as the composition of two functions.

■ 3. Write $f(x) = 5(2\sqrt[3]{x})^2 - 8$ as the composition of two functions.

■ 4. Write $f(x)$ as the composition of two functions.

$$f(x) = \frac{\frac{1}{x+1}}{\frac{1}{x+1} - 1}$$

■ 5. Write $f(x) = 5(2x + 3)^4 + 3(2x + 3)^2 - 7$ as the composition of two functions.

■ 6. Write $f(x)$ as the composition of two functions.

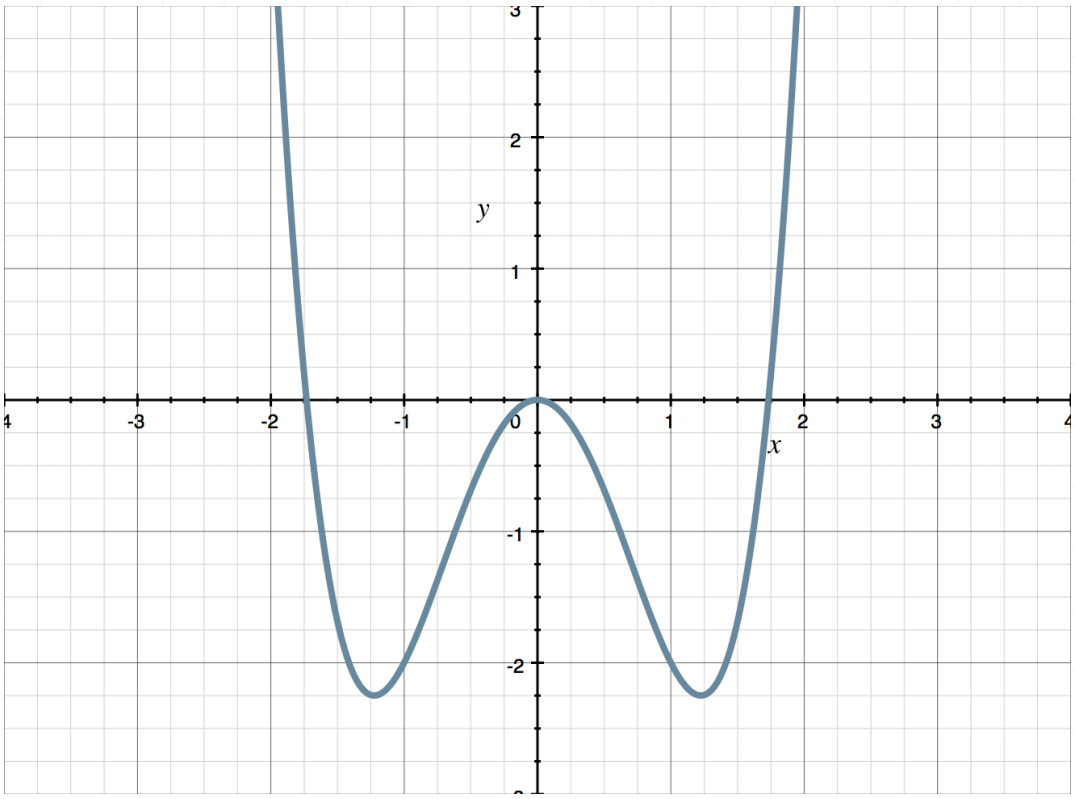
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$$f(x) = \frac{4}{x^2 - 7}$$

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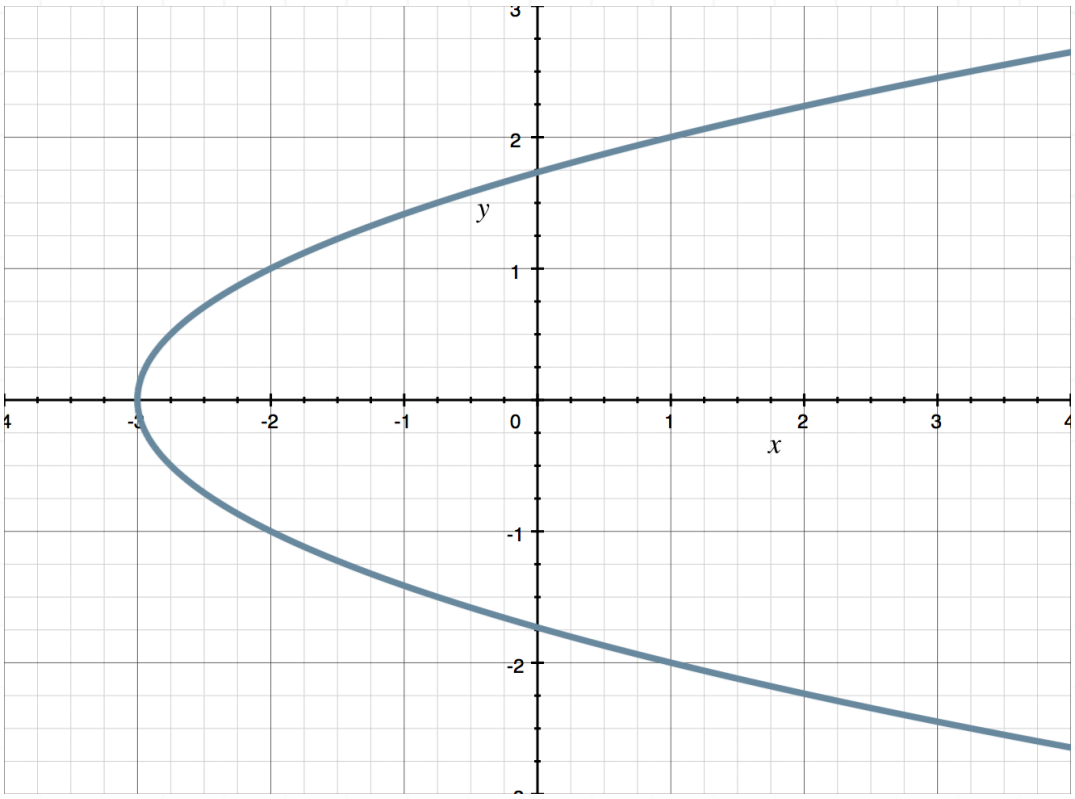
ONE-TO-ONE FUNCTIONS AND THE HORIZONTAL LINE TEST

1. Does the graph represent a one-to-one function?

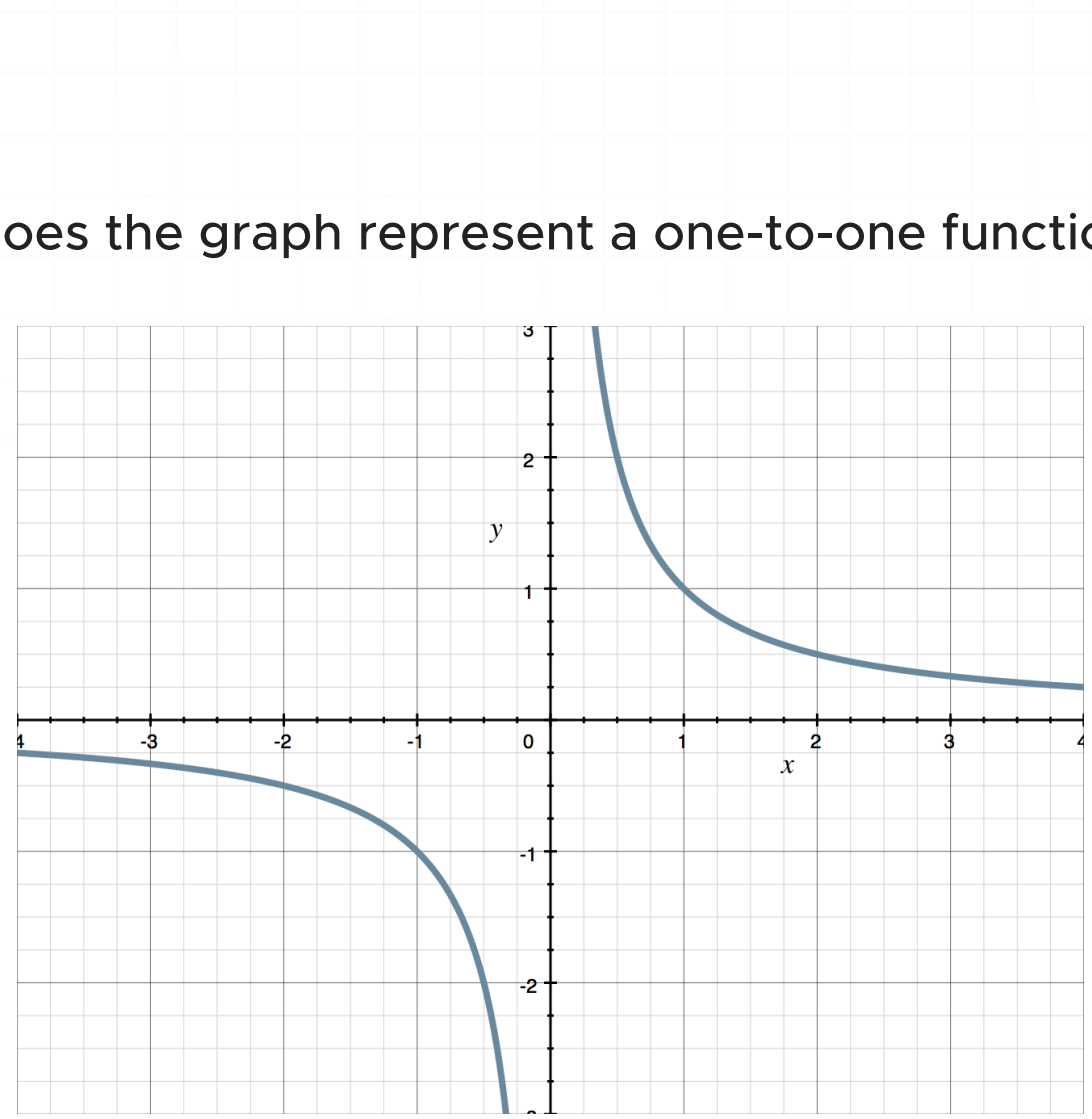


2. Does the graph represent a one-to-one function?

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3. Does the graph represent a one-to-one function?



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■ 4. Show that the function is one-to-one by showing that $f(a) = f(b)$ leads to $a = b$.

$$f(x) = 3x - 4$$

■ 5. Show that the function is one-to-one by showing that $f(a) = f(b)$ leads to $a = b$.

$$f(x) = \frac{x + 1}{x - 5}$$

■ 6. Show that the function is not one-to-one by showing that $f(a) = f(b)$ does not lead to $a = b$.

$$f(x) = (x + 3)(x - 2)$$

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INVERSE FUNCTIONS

■ 1. What is the inverse of the function?

$$f(x) = \frac{1}{2}x - 3$$

■ 2. What is the inverse of the function?

$$f(x) = -4x + 5$$

■ 3. What is the inverse of the function?

$$f(x) = \frac{2x}{x - 5}$$

■ 4. What is the inverse of the function?

$$f(x) = \frac{1}{x} + 3$$

■ 5. What is the inverse of the function?

$$f(x) = -\frac{3}{x - 2} - 4$$

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6. What is the inverse of the function?

$$f(x) = \frac{x-2}{x+3}$$

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FINDING A FUNCTION FROM ITS INVERSE

■ 1. Find $f(x)$ if $f^{-1}(x)$ is a linear function.

$$f^{-1}(1) = -2$$

$$f^{-1}(-3) = -1$$

■ 2. Find $f(x)$ if $f^{-1}(x)$ is a linear function.

$$f^{-1}(0) = 3$$

$$f^{-1}(-2) = 1$$

■ 3. Find $f(x)$ if $f^{-1}(x)$ is a linear function.

$$f^{-1}(2) = 5$$

$$f^{-1}(4) = 9$$

■ 4. Find $f(x)$ if $f^{-1}(x)$ is a linear function.

$$f^{-1}(-4) = 7$$

$$f^{-1}(-1) = 14$$

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■ 5. Find $f(x)$ if $f^{-1}(x)$ is a linear function.

$f^{-1}(5) = -4$

$f^{-1}(10) = -12$

■ 6. Find $f(x)$ if $f^{-1}(x)$ is a linear function.

$f^{-1}(1) = 3$

$f^{-1}(2) = 6$

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WHAT IS A LOGARITHM?

■ 1. How would you read the logarithmic equation out loud?

$$\log_7 57 = y$$

■ 2. Rewrite the equation in logarithmic form.

$$64^{\frac{1}{2}} = 8$$

■ 3. Rewrite the equation in exponential form.

$$\log_u \frac{19}{20} = v$$

■ 4. Use the general log rule to solve for x.

$$\log_5(125) = x$$

■ 5. Use the general log rule to solve the logarithm.

$$\log_8 64 = x$$

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6. Write the logarithm that answers this question: To what power do you have to raise 289 in order to get 17?

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COMMON BASES AND RESTRICTED VALUES

- 1. Is there something wrong with the logarithm? If so, what is it?

$$\log_{-5}(8)$$

- 2. Is there something wrong with the logarithm? If so, what is it?

$$\log_5(-8)$$

- 3. Is the following statement true or false? Why?

$$\log 5 = \log_{10} 5$$

- 4. How else can you write $\log_e 7$?

- 5. Solve the logarithm for x.

$$\log(10{,}000) = x$$

- 6. Solve the logarithm for x.

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$\ln(e^5) = x$

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EVALUATING LOGS

1. Find the exact value of the logarithm.

$$\log_{\frac{1}{3}} \frac{1}{27}$$

2. Find the exact value of the logarithm.

$$\log_9 \frac{1}{81}$$

3. Find the value given by the log.

$$\log_{7,776} 6$$

4. Find the exact value of the logarithm.

$$\log_{343} 7$$

5. Find the exact value of the logarithm.

$$\log_{\frac{1}{4}}(1{,}024)$$

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6. Find the exact value of the logarithm.

$$\log_{64} \frac{1}{4}$$

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LAWS OF LOGARITHMS

■ 1. Write the expression as a single logarithm. Solve if possible.

$$\log_2 2 + \log_2 4$$

■ 2. Write the expression as a single logarithm. Solve if possible.

$$\log_3 216 - \log_3 24$$

■ 3. Write the expression as a single logarithm. Solve if possible.

$$\log_4 10 - 3\log_4 2$$

■ 4. Write the expression as a single logarithm. Solve if possible.

$$2\log_7 4 + 3\log_7 5$$

■ 5. Simplify the logarithmic expression.

$$\log_2\left(\frac{x^4y^2}{z^3}\right)$$

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6. Simplify the logarithmic expression.

$$\log_9\left(\frac{9x^3}{2y^4z}\right)$$

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LAWS OF NATURAL LOGS

■ 1. Condense the expression into a single logarithm.

$$2\ln 2 - 3\ln 4$$

■ 2. Condense the expression into a single logarithm.

$$\frac{5\ln 2}{4}$$

■ 3. Condense the expression into a single logarithm.

$$3(\ln 2x - \ln 5) + (\ln 4 - \ln 3y)$$

■ 4. Expand the logarithm.

$$\ln (3 \cdot 5)^4$$

■ 5. Expand the logarithm.

$$\ln \left(\frac{4}{5}\right)^2$$

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6. Expand the logarithm.

$$\ln(3xy^2)$$

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SOLVING LOGARITHMIC EQUATIONS

■ 1. Solve the equation.

$$2\log_b x = \log_b 49$$

■ 2. Solve the equation.

$$\log_{12} x = \frac{3}{2}\log_{12} 16$$

■ 3. Solve the equation.

$$\log_5(x-1) + \log_5(x+5) = \log_5 7$$

■ 4. Solve the equation.

$$\log_9(-4x) - \log_9 12 = \log_9 5$$

■ 5. Solve the equation.

$$\log_a 2 + \log_a 4 = \log_a(x+2)$$

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6. Solve the equation.

$$\log_4(x+5) - \log_4(x-2) = \log_4 3$$

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CHANGE OF BASE

■ 1. Find the value of the expression to four decimal places.

$$\log_3 6$$

■ 2. Find the value of the expression to four decimal places.

$$\frac{\log 25}{\log 5} - \frac{\log 25}{\log 125}$$

■ 3. Find the approximate value to the expression to the nearest ten thousandth.

$$\frac{\log 18}{\log 2} - \frac{\log 9}{\log 2}$$

■ 4. Write the log expression in terms of natural logs.

$$\log_4 6.7$$

■ 5. Find the exact value of the logarithmic expression.

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$$\frac{\ln(16{,}807)}{\ln(7)}$$

6. Use logarithms to solve the equation to the nearest ten thousandth.

$$8 \cdot 6^{3x} = 4{,}104$$

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GRAPHING EXPONENTIAL FUNCTIONS

■ 1. Will the function have a vertical or horizontal asymptote? Where is it located? What is the end behavior of the function?

$$y = \left(\frac{1}{3}\right)^{x-2} + 3$$

■ 2. Will the function have a vertical or horizontal asymptote? Where is it located?

$$x = -5^{y+1} - 8$$

■ 3. Graph the exponential function.

$$f(x) = \left(\frac{4}{5}\right)^{x}$$

■ 4. What is the x-intercept of the function?

$$x = \left(\frac{11}{8}\right)^{y} - 4$$

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■ 5. Sketch the graph of the function by finding the y-intercept and figuring out the function's end behavior.

$$y = -2^{x+1} - 5$$

■ 6. Sketch the graph of the function using the x-intercept and the end behavior.

$$x = \left(\frac{1}{4}\right)^{y+2} - 2$$

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GRAPHING TRANSFORMATIONS OF EXPONENTIAL FUNCTIONS

■ 1. Sketch the graph of the function.

$$y = 3^{x-2} - 3$$

■ 2. Sketch the graph of the function.

$$y = -2(2^{x+3}) + 1$$

■ 3. Graph the exponential function.

$$f(x) = \left(\frac{1}{2}\right)^{-x-2} - 5$$

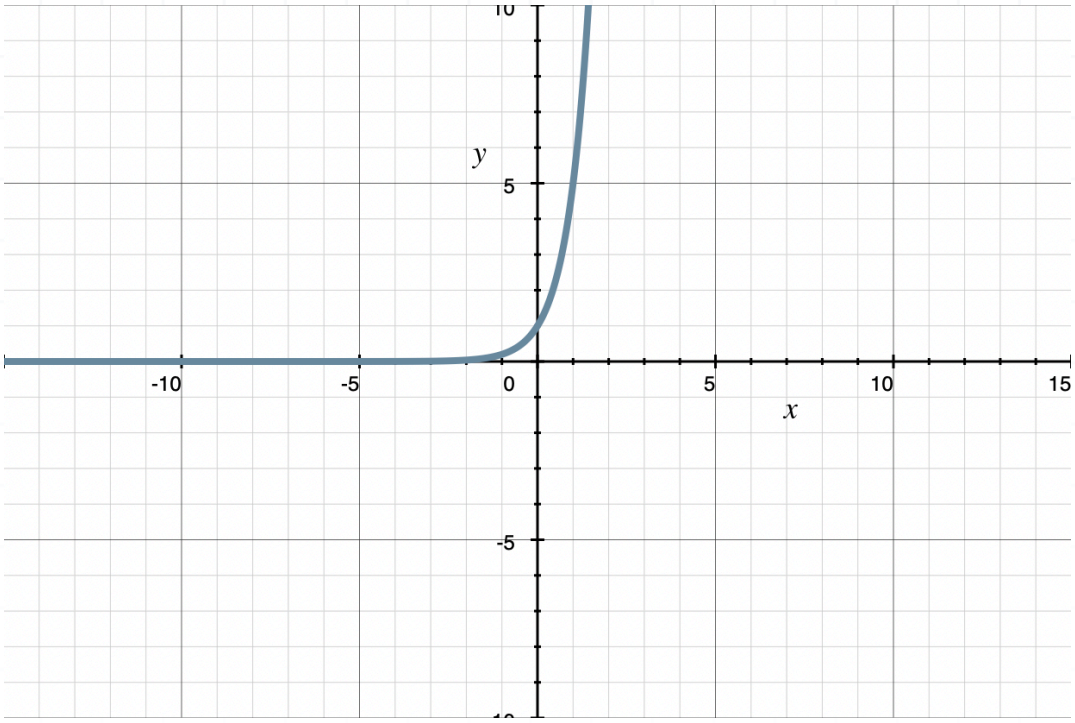
■ 4. Write the equation for the function that results from applying the list of transformations below to the parent function $y = 6^x$. The list of transformations is not necessarily given in the order in which they should be applied.

- Vertically stretch by a factor of 2
- Vertical shift 4 units up and horizontal shift 3 units to the right
- Reflection across the x-axis

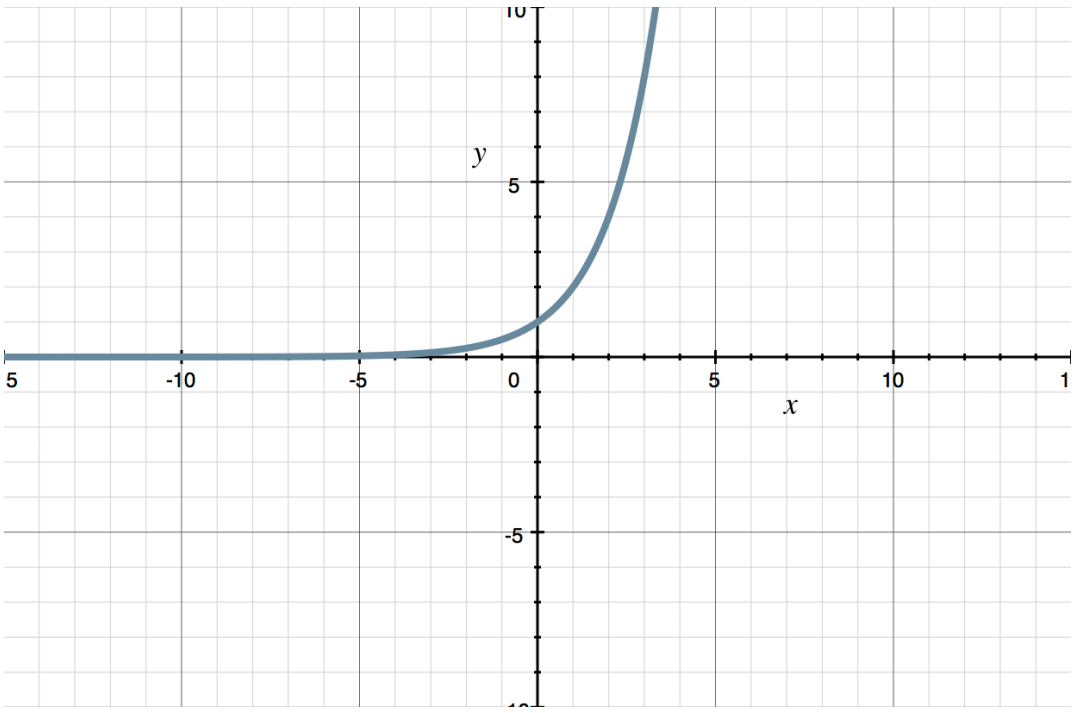
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- Reflection across the y-axis

■ 5. Given the graph of $y = 5^x$, use transformations to graph $y = -5^{x+1} - 2$.



■ 6. Given the graph of $y = 2^x$, use transformations to graph $y = 2^{x-1} + 4$.



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THE GENERAL LOG RULE

■ 1. Write the inverse of the log function.

$$\log y = 2x$$

■ 2. Write the inverse of the log function.

$$\ln y = x$$

■ 3. The table shows points that satisfy an exponential function. Write a set of four points that will satisfy its inverse.

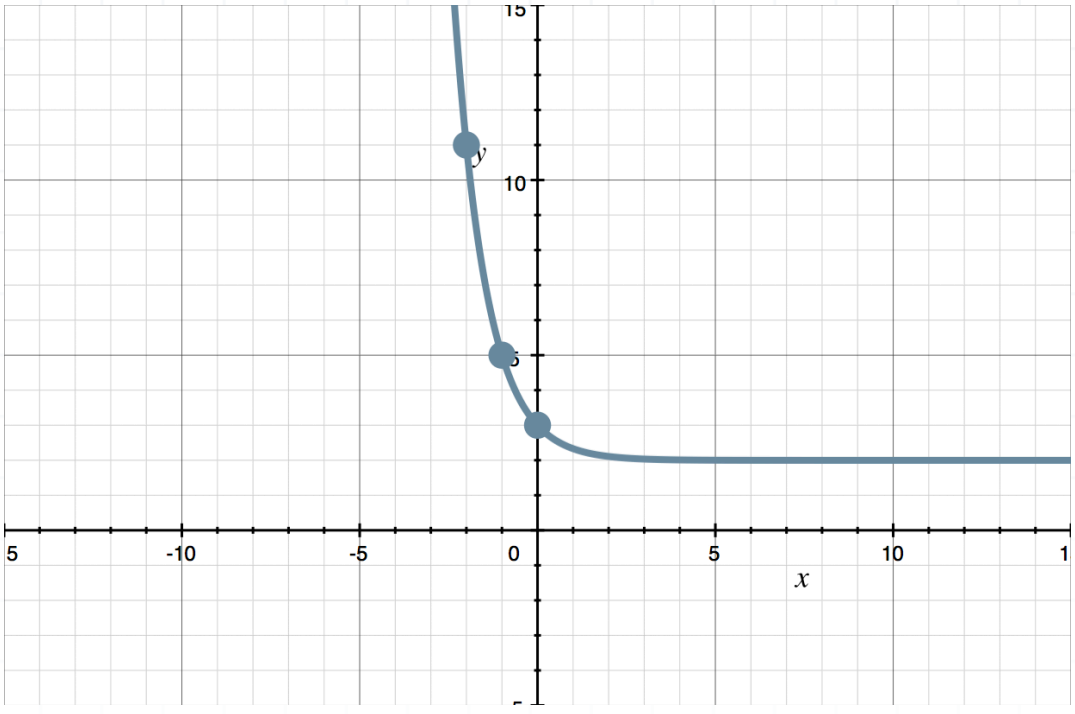
x	1	2	3	4
$y=a^x$	1.5	2.25	3.375	5.0625

■ 4. The table shows points that satisfy a logarithmic function. Write a set of four points that will satisfy its inverse.

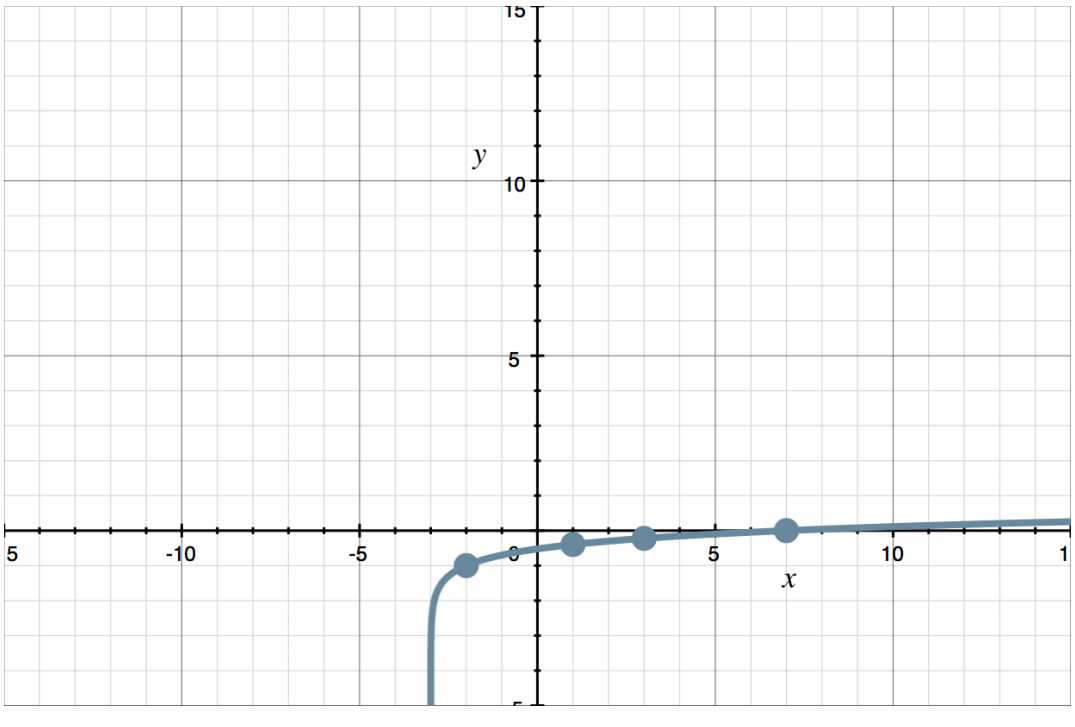
x	10	100	1,000	10,000
y=log(x)	1	2	3	4

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■ 5. The graph shown passes through $(-2,11)$, $(-1,5)$, and $(0,3)$. Sketch the graph of its inverse.



■ 6. The graph shown passes through $(-2,-1)$, $(1,-0.3979)$, $(3,-0.2218)$, and $(7,0)$. Sketch the graph of its inverse.



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GRAPHING LOG FUNCTIONS

■ 1. Will the function have a vertical or horizontal asymptote? Where is it located?

$$y = \log_2(x - 1)$$

■ 2. Graph the function $y = \log_6(x)$.

■ 3. Sketch the graph of the function using intercepts and end behavior.

$$y = 5\log_2(x + 4)$$

■ 4. Graph the function $y = \ln(x)$.

■ 5. Sketch the graph of the function by making a table of values.

$$y = \log_5 x$$

■ 6. Sketch the graph of the function using intercepts and end behavior.

$$x = 2\log_3(y + 1)$$

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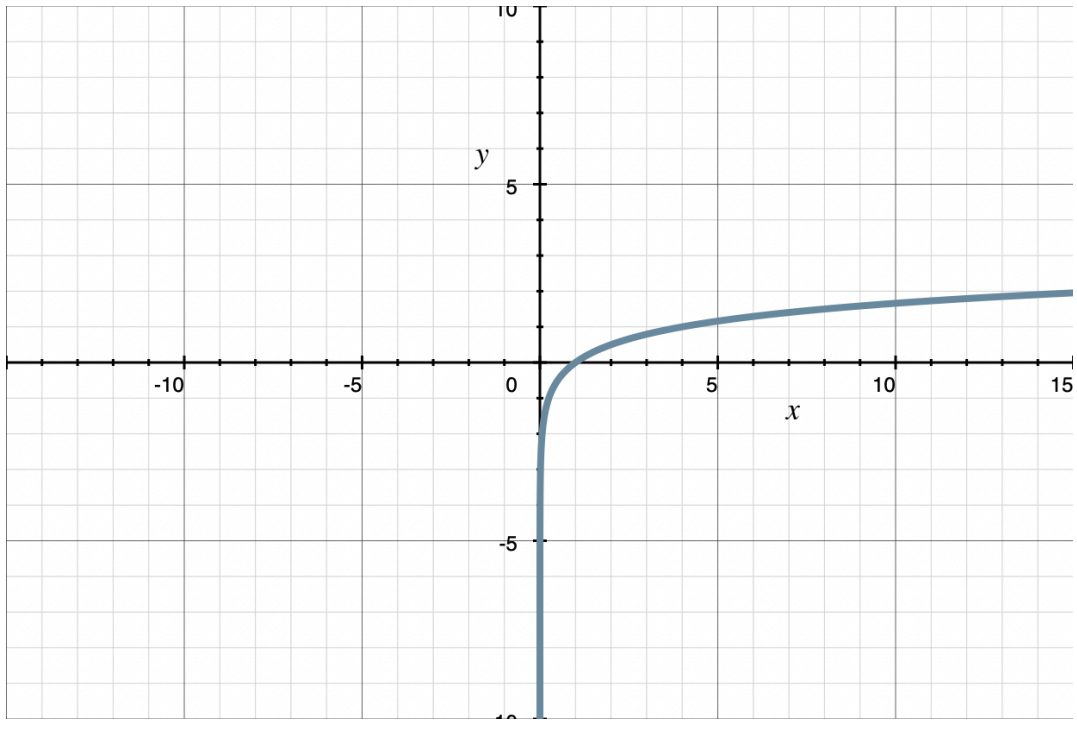
GRAPHING TRANSFORMATIONS OF LOG FUNCTIONS

■ 1. Use transformations to sketch the graph of $y = -3\log_7(x-2)+4$.

■ 2. Will the function have a vertical or horizontal asymptote? Where is it located?

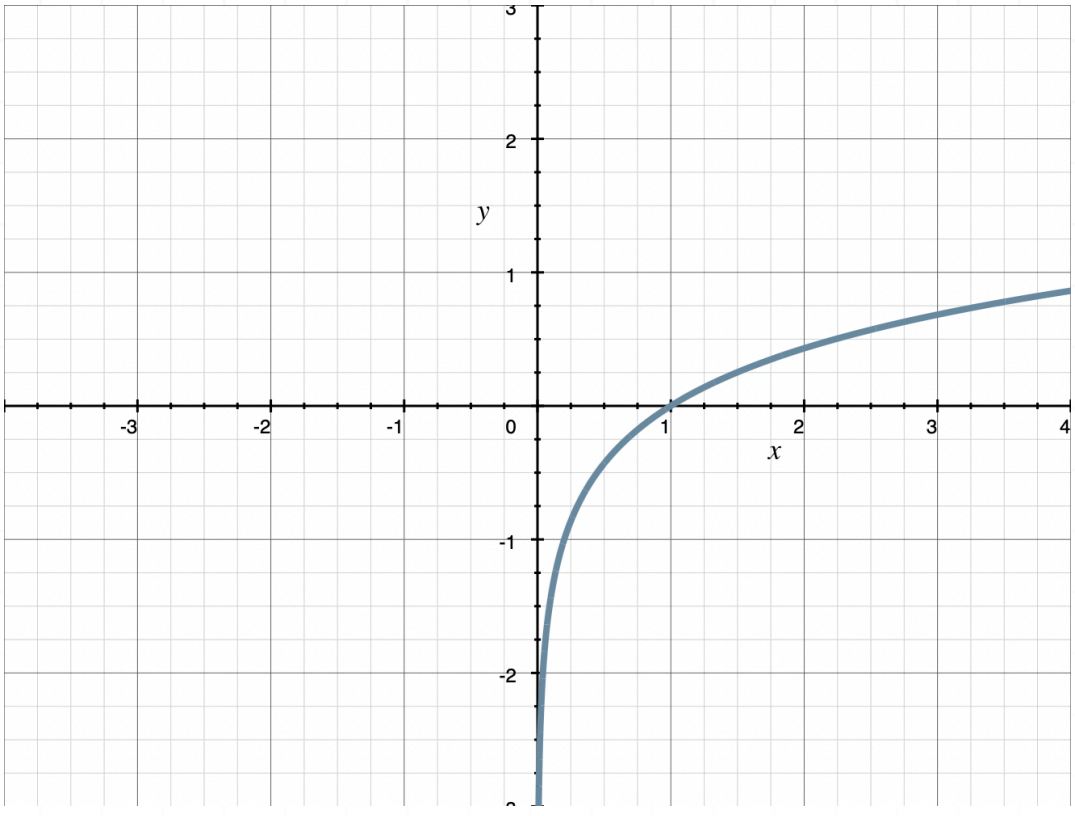
$$y = \log_4(2x) + \log_4(3)$$

■ 3. Given the graph of $y = \log_4(x)$, use transformations to sketch the graph of $y = \log_4(x-2)-3$.



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■ 4. Given the graph of $y = \log_5(x)$, use transformations to sketch the graph of $y = -\log_5(x + 1) - 2$.



■ 5. Use transformations to sketch the graph of $y = \log_4(x + 8)$.

■ 6. Use transformations to sketch the graph of $y = \log(5x) - 3$.

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